

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
	(e)			award (1) for appropriate unit conversion (kJ mol^{-1} to J mol^{-1} or reverse for R) rearrangement of equation gives $A = k \div e^{-(79333/RT)}$ (1) $A = 1.864 \times 10^{13}$ (1) $k = 0.167$ (1)	1		1 1 1	4	4	
	(f)			any three for (1) each - indicator colour may be difficult to see due to coloured compounds / the vanadium compounds change colour during titration - colour change of transition metals can be used in place of an indicator - for colorimetry find a wavelength OR frequency of light absorbed <u>only by the VO^{2+} / Ce^{4+}</u> - end point is when there is no absorbance at VO^{2+} frequency / <u>Ce^{4+}</u> remains in solution		2	1	3		3
				Question 12 total	5	3	9	17	6	6